TAFWELA

Non-Functional Requirements

(DPEI) | 2026

1. Purpose

This document defines the Non-Functional Requirements (NFRs) for the TAFWELA application. NFRs describe how the system performs its functions — covering performance, security, usability, reliability, maintainability, compatibility, and scalability constraints.

2. NFR Summary by Category

Category	# of NFRs	Key Focus
Performance	**5**	Speed, responsiveness, memory usage
Security	**5**	Data protection, role access, encryption
Usability	**5**	UI consistency, accessibility, UX quality
Reliability	**5**	Uptime, crash rate, offline handling
Maintainability	**4**	Code quality, testing, documentation
Compatibility	**3**	Android versions, screen sizes, orientations
Scalability	**2**	Future growth, modular architecture
TOTAL	**29**	

3. Non-Functional Requirements Register

3.1 Performance

ID	Requirement	Metric / Target	Rationale
NFR-01	The app shall load and display the map with station markers within 2 seconds on a 4G connection.	**Load time ≤ 2 seconds measured from app launch to map ready**	Drivers expect instant visibility — delays cause abandonment.
NFR-02	Congestion status updates shall be reflected on all client maps within 1 second of a staff update.	**Firestore listener latency ≤ 1 second**	Real-time accuracy is the core value proposition of TAFWELA.
NFR-03	The GPS location shall be detected within 3 seconds on first launch.	**Geolocator response time ≤ 3 seconds**	Drivers should not wait for their location to be determined.
NFR-04	The station list view shall render within 1.5 seconds for up to 50 stations.	**List render time ≤ 1.5 seconds with 50 items**	Ensures smooth experience even with a large number of nearby stations.
NFR-05	The app shall not consume more than 80MB of RAM during normal usage.	**Memory usage ≤ 80MB monitored via Flutter DevTools**	Prevents performance degradation on low-end Android devices common in Egypt.

3.2 Security

ID	Requirement	Metric / Target	Rationale
NFR-06	All user data shall be transmitted over HTTPS using Firebase's built-in TLS encryption.	**100% of API calls use HTTPS / TLS**	Protects driver and staff data from interception.
NFR-07	Firestore Security Rules shall prevent unauthorized read/write access to any collection.	**0 unauthorized accesses allowed; rules tested with Firebase Emulator**	Ensures data integrity and role-based access control.
NFR-08	Staff Panel shall only be accessible to accounts with the 'staff' role verified via Firestore.	**Role verification enforced on every Staff Panel screen entry**	Prevents regular drivers from accessing or modifying station status.
NFR-09	Passwords shall be stored and managed exclusively by Firebase Auth — never stored locally.	**No password stored in local storage or device memory**	Firebase Auth handles all credential management securely.
NFR-10	The app shall automatically log out inactive staff sessions after 24 hours.	**Session timeout ≤ 24 hours for staff accounts**	Reduces risk of unauthorized access on shared devices.

3.3 Usability

ID	Requirement	Metric / Target	Rationale
NFR-11	The app UI shall be consistent with Material Design 3 guidelines throughout all screens.	**Consistent use of Material Design 3 components**	Provides a familiar, intuitive interface for Egyptian users.
NFR-12	All interactive elements (buttons, markers, inputs) shall have a minimum tap target size of 48×48dp.	**Tap targets ≥ 48×48dp on all interactive elements**	Reduces accidental taps, especially while driving or in poor lighting.
NFR-13	Congestion status shall be communicated using both color AND text/icon to support color-blind users.	**Color + text label on all congestion indicators**	Ensures accessibility for color-blind users (affects ~8% of males).
NFR-14	Error messages shall be descriptive, actionable, and displayed in plain language.	**All error states have user-friendly messages**	Helps drivers and staff understand what went wrong and what to do next.
NFR-15	The onboarding flow shall be completable in under 30 seconds.	**Onboarding completion time ≤ 30 seconds**	Minimizes friction for new users getting started quickly.

3.4 Reliability

ID	Requirement	Metric / Target	Rationale
NFR-16	The app shall have a crash rate of less than 1% during the testing phase.	**Crash rate < 1% across all test sessions**	High crash rate directly damages user trust and project evaluation.
NFR-17	Firebase Firestore availability shall meet a 99.5% uptime target during the project period.	**Firebase uptime ≥ 99.5% (monitored via Firebase Console)**	Backend availability is critical for real-time data delivery.
NFR-18	The app shall handle network disconnection gracefully by showing a clear offline message.	**Offline state detected and displayed within 2 seconds**	Egyptian network connectivity can be unreliable in some areas.
NFR-19	All Firestore write operations shall include error handling with retry logic.	**100% of write operations have try/catch and retry**	Prevents silent data loss on flaky connections.
NFR-20	The app shall retain user session between app restarts without requiring re-login.	**Session persists across app restarts via Firebase Auth persistence**	Improves user experience by avoiding repeated login prompts.

3.5 Maintainability

ID	Requirement	Metric / Target	Rationale
NFR-21	The codebase shall follow Flutter Clean Architecture principles with clear separation of Data, Domain, and Presentation layers.	**Clean Architecture layers enforced in all modules**	Ensures code is testable, scalable, and easy to modify.
NFR-22	All public methods and classes shall include Dart doc comments.	**Dart doc coverage ≥ 70% of public APIs**	Helps team members understand and maintain each other's code.
NFR-23	Unit test coverage shall cover at least 60% of core business logic.	**flutter test --coverage report shows ≥ 60%**	Reduces regression risk during integration and bug fixes.
NFR-24	The project shall use a consistent naming convention: camelCase for variables, PascalCase for classes, snake_case for files.	**Naming convention enforced via code review**	Ensures readability and consistency across all 6 team members' code.

3.6 Compatibility

ID	Requirement	Metric / Target	Rationale
NFR-25	The app shall support Android versions 6.0 (API 23) and above.	**Tested on Android 6, 9, 11, 13**	Ensures compatibility with the wide range of Android devices used in Egypt.
NFR-26	The app UI shall be responsive and render correctly on screen sizes from 5 to 6.7 inches.	**Tested on small (5"), medium (6"), and large (6.7") screens**	Egyptian users use a wide variety of phone sizes.
NFR-27	The app shall function correctly on both portrait and landscape orientations.	**No layout breakage in either orientation**	Prevents display issues in different usage contexts.

3.7 Scalability

ID	Requirement	Metric / Target	Rationale
NFR-28	The Firestore database schema shall support up to 10,000 fuel stations without requiring structural changes.	**Schema designed for ≥ 10,000 station documents**	Allows future expansion beyond the initial 20 seed stations.
NFR-29	The app architecture shall allow new features to be added as independent modules without modifying existing code.	**New modules add without breaking existing features (Open/Closed Principle)**	Supports future v2 features like payment or ratings.